Introduction — The Problem of Measure

Shekhar Rathod

Consider the set of all real numbers, $\mathbb{R}$. **Our goal is to assign a *measure* to every subset of $\mathbb{R}$.** Without defining this term rigorously, we have a certain intuitive notion of how this measure on $\mathbb{R}$ corresponds to length. For instance, it makes sense to say that the measure of the interval (a, b) is $b-a$, and that translating this interval by a certain quantity, say x, should not result in a change in its measure; that is, the measure of $(a+x, b+x)$ should remain $b-a$ for all $x \in \mathbb{R}$. We would like to extend this notion from open intervals like (a, b) to all subsets of $\mathbb{R}$.

Our objective, then, is to find a function λ that satisfies the following:

(P1) To every subset of $\mathbb{R}$, λ should assign some nonnegative value, possibly $+\infty$. Hence, it should be a function from $\mathscr{P}(\mathbb{R})$ to $[0, +\infty]$, i.e., $\lambda : \mathscr{P}(\mathbb{R}) \to [0, +\infty]$.

(P2) $\lambda((a, b)) = b - a$.

(P3) $\lambda(A + x) = \lambda(A)$, where $A + x := \{a + x : a \in A\}$.

(P4) Let $A_1, A_2, \ldots$ be a sequence of pairwise disjoint subsets of $\mathbb{R}$. Then, $\lambda(\bigsqcup_{j \geq 1} A_j) = \sum_{j \geq 1} \lambda(A_j)$.

We now show that such a function cannot exist.

Lemma. *Let E and F be subsets of $\mathbb{R}$ such that $E \subseteq F$. Then, $\lambda(E) \leq \lambda(F)$.*

Proof. We first show that $\lambda(\varnothing) = 0$. Set $A_j = \varnothing$ for all $j \geq 1$ in (P4). Then, $\lambda(\varnothing) = \sum \left(\bigsqcup_{j \geq 1} \varnothing\right) = \sum_{j \geq 1} \lambda(\varnothing)$, which is possible only if $\lambda(\varnothing) = 0$.

If $E \subseteq F$, then $F = E \cup (F \setminus E)$, and E and $F \setminus E$ are disjoint. Setting $A_1 = E$, $A_2 = F \setminus E$, and $A_j = \varnothing$ for all $j \geq 3$ in (P4), we get $\lambda(F) = \lambda(E) + \lambda(F \setminus E) \geq \lambda(E)$. **QED**

Theorem. *A function λ satisfying (P1) to (P4) does not exist.*

Proof. Suppose, for contradiction, that such a function λ does exist.

Define a relation $\sim$ such that $a \sim b$ if and only if $a - b \in \mathbb{Q}$.

Clearly, $\sim$ is an equivalence relation. Define $[x] := \{y \in \mathbb{R} : x - y \in \mathbb{Q}\}$. Then, $[x]$ is an equivalence class. Observe that $[x] = \{x + q : q \in \mathbb{Q}\}$. Hence, there is a bijection $q \mapsto x + q$ from $\mathbb{Q}$ to $[x]$, and so $[x]$ is a countable set.

Let Λ be the collection of all equivalence classes. Then, Λ is uncountably infinite. For if Λ were to be countably infinite, then, since each equivalence class is also countable, the union of all the equivalence classes would be a countable union of countable sets, which would also be a countable set. However, the union of all equivalence classes is $\mathbb{R}$, since every real number belongs to some equivalence class, and $\mathbb{R}$ is uncountable, which is a contradiction.

By the Axiom of Choice, we can choose one element from each equivalence class. Call the resulting set Ω, and we may assume that $\Omega \subseteq (0,1)$, for we can choose an element from each equivalence class that is in $(0,1)$.

We first show that $\lambda(\Omega) = 0$.

Let $q_1, q_2 \in \mathbb{Q}$, $q_1 \neq q_2$. We now show that $(\Omega + q_1) \cap (\Omega + q_2) = \varnothing$.

Let $x \in (\Omega + q_1) \cap (\Omega + q_2)$. Then, $x \in \Omega + q_1$ and $x \in \Omega + q_2$, that is, $x = \alpha + q_1$ and $x = \beta + q_2$ for some $\alpha, \beta \in \Omega$. Hence, $\alpha + q_1 = \beta + q_2$, or, $\alpha - \beta = q_2 - q_1 \in \mathbb{Q}$. Therefore, $\alpha \sim \beta$, and so α and β belong to the same equivalence classes. However, both α and β are in Ω, which contains exactly one element from each equivalence class, by construction. This implies that $\alpha = \beta$, and so $q_1 = q_2$, which contradicts that $q_1 \neq q_2$, proving that $(\Omega + q_1) \cap (\Omega + q_2) = \varnothing$.

Now consider $-1 < q < 1$, $q \in \mathbb{Q}$. Since $\Omega \subseteq (0,1)$,

$$\bigsqcup_{\substack{-1<q<1\\ q\in\mathbb{Q}}} (\Omega + q) \subseteq (-1, 2),$$

and by (P4) along with the lemma,

$$\sum_{\substack{-1<q<1\\ q\in\mathbb{Q}}} \lambda(\Omega + q) \leq \lambda((-1,2)) = 3.$$

But by (P3), $\lambda(\Omega + q) = \lambda(\Omega)$, and so

$$\sum_{\substack{-1<q<1\\ q\in\mathbb{Q}}} \lambda(\Omega) \leq \lambda((-1,2)) = 3,$$

which is only possible if $\lambda(\Omega) = 0$, since there are infinitely many rationals between -1 and 1. Hence, $\lambda(\Omega) = 0$.

We now show:

$$(0,1) \subseteq \bigsqcup_{\substack{-1<q<1\\ q\in\mathbb{Q}}} (\Omega + q).$$

Suppose $x \in (0,1)$. Let $x_0 = [x] \cap \Omega$. Then, $x \sim x_0$, i.e., $x - x_0 = q \in \mathbb{Q}$. Now, $x_0 \in (0,1)$ (for $x_0 \in \Omega \subseteq (0,1)$), and $x \in (0,1)$, which implies that $q = x - x_0 \in (-1,1)$. Thus, $x = x_0 + q \in \Omega + q$ for some rational q such that $-1 < q < 1$. Hence, $x \in \bigsqcup_{\substack{-1<q<1\\ q\in\mathbb{Q}}} (\Omega + q)$.

By the lemma,

$$\lambda((0,1)) \leq \lambda\left(\bigsqcup_{\substack{-1<q<1\\ q\in\mathbb{Q}}} (\Omega + q) \right),$$

i.e.,

$$1 \leq \sum_{\substack{-1<q<1\\ q\in\mathbb{Q}}} \lambda(\Omega + q),$$

which is a contradiction, for the sum on the right is 0. **QED**